

Partial Lower Bounds for Stochastic Acceleration under the Growth Condition

Draft note

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Abstract

Chen, Na, and Kolar asked for lower-bound evidence clarifying whether the small multiplicative-noise assumptions in their accelerated SGD analysis under the growth condition are necessary. This note gives a self-contained lower-bound construction for three constant-parameter methods: Nesterov’s accelerated method (NAM), the robust momentum method (RMM), and constant-ratio implicit dual averaging (iDAM+). The construction uses only one-dimensional strongly convex quadratics and an unbiased multiplicative-noise oracle satisfying the growth condition with zero additive noise.

The result is *partial progress, not a solution of the sharp-threshold problem*. It proves that uniform accelerated mean-square bias contraction of order $1 - \Theta(\kappa^{-1/2})$ is impossible for NAM when $\delta \geq 2$, for RMM when $\delta \geq 1$, and for constant-ratio iDAM+ when $\delta \geq 2$. The proof leaves open the intervals $1 \leq \delta < 2$ for NAM and iDAM+, and $1/4 \leq \delta < 1$ for RMM, relative to the sufficient regimes proved by Chen–Na–Kolar.

1 Introduction and status of the result

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be m -strongly convex and L -smooth, with condition number

$$\kappa = \frac{L}{m}.$$

A stochastic first-order oracle returns

$$g(x) = \nabla f(x) + \varepsilon(x), \quad \mathbb{E}[\varepsilon(x) \mid x] = 0,$$

and is said to satisfy the growth condition if

$$\mathbb{E}\|\varepsilon(x)\|^2 \leq \delta \|\nabla f(x)\|^2 + \sigma^2. \tag{1}$$

Chen, Na, and Kolar [1] analyze several accelerated stochastic-gradient methods under (1). Their sufficient results show accelerated bias contraction for NAM and iDAM+ under a small-noise condition $\delta < 1$, for RMM under $\delta < 1/4$, and for DAM+ under arbitrary δ . They explicitly note that they do not provide lower bounds for NAM, RMM, or iDAM+, and that the necessity and sharpness of the small- δ conditions are unclear. The open-problem record [2] asks for sharp necessary conditions on δ for acceleration of NAM, RMM, and iDAM+.

This note does *not* determine those sharp necessary conditions. It proves only the following impossibility ranges.

Method	Sufficient acceleration in [1]	Lower bound proved here	Gap left open
NAM	$\delta < 1$	no acceleration for $\delta \geq 2$	$1 \leq \delta < 2$
RMM	$\delta < 1/4$	no acceleration for $\delta \geq 1$	$1/4 \leq \delta < 1$
iDAM+	$\delta < 1$	no acceleration for $\delta \geq 2$	$1 \leq \delta < 2$

Thus the contribution is a rigorous adversarial construction showing that sufficiently large multiplicative noise destroys constant-parameter acceleration for these three schemes. It should be read as partial progress toward the lower-bound part of the problem, not as a complete resolution of the authors’ sharp-threshold question.

1.1 Meaning of “accelerated” in this note

For a method with state S_k and distinguished output iterate y_k , I use the following standard uniform-bias meaning. A constant-parameter tuning has an accelerated mean-square bias contraction if there exist constants $c > 0$, $C < \infty$, and κ_0 , independent of κ , such that for every $\kappa \geq \kappa_0$, every admissible objective and oracle with $\sigma^2 = 0$, and every deterministic initial state,

$$\mathbb{E}\|y_k - y_\star\|^2 \leq C \left(1 - \frac{c}{\sqrt{\kappa}}\right)^k \|S_0 - S_\star\|^2. \quad (2)$$

On the one-dimensional quadratic instances below, objective bias is simply a curvature multiple of $\mathbb{E}y_k^2$, so a lower bound on squared-iterate contraction is also a lower bound on objective-bias contraction.

The parameter ranges are the usual constant-parameter ranges for the methods considered here:

$$\eta > 0, \quad 0 \leq \beta < 1$$

for NAM and RMM, and a constant ratio

$$0 < \tau = \frac{a_k}{A_k} < 1, \quad \frac{A_{k-1}}{A_k} = 1 - \tau$$

for iDAM+. The iDAM+ statement is therefore a statement about constant-ratio iDAM+, not about all possible time-varying iDAM+ schedules.

2 The hard oracle and two elementary lemmas

For $\lambda \in [m, L]$, define the one-dimensional quadratic

$$f_\lambda(x) = \frac{\lambda}{2}x^2, \quad x_\star = 0.$$

Let $(s_k)_{k \geq 0}$ be independent Rademacher signs, and set

$$\zeta_k = 1 + \sqrt{\delta} s_k, \quad g_k = \lambda x_k \zeta_k.$$

Then

$$\mathbb{E}[g_k \mid x_k] = \lambda x_k = \nabla f_\lambda(x_k),$$

and, writing $\varepsilon_k = g_k - \nabla f_\lambda(x_k)$,

$$\mathbb{E}[\varepsilon_k^2 \mid x_k] = \delta \lambda^2 x_k^2 = \delta |\nabla f_\lambda(x_k)|^2.$$

Therefore the oracle satisfies (1) with additive noise $\sigma^2 = 0$. The construction uses no boundedness, sub-Gaussianity, or nonnegativity assumption on the noisy multiplier ζ_k .

Each method below becomes a homogeneous linear random recursion

$$S_{k+1} = A(\zeta_k)S_k.$$

Its mean transition is

$$T = \mathbb{E}A(\zeta_k),$$

and its second-moment transition is the matrix induced by

$$\mathbb{E}[S_{k+1} \otimes S_{k+1}] = \mathbb{E}[A(\zeta_k) \otimes A(\zeta_k)] \mathbb{E}[S_k \otimes S_k].$$

The latter matrix will be denoted by M .

Lemma 2.1 (A mean mode lower-bounds mean-square contraction). *Suppose T has an eigenvalue μ with an eigenvector whose output-coordinate component is nonzero. Then no estimate of the form (2) can hold with a per-iteration contraction factor strictly smaller than $|\mu|^2$. In particular, any asymptotic mean-square contraction factor for the output is at least $\rho(T)^2$, provided a spectral-radius eigenmode is visible in the output coordinate.*

Proof. Take a deterministic initial state in the real span of the real and imaginary parts of the eigenvector. The unbiasedness of the oracle gives

$$\mathbb{E}S_k = T^k S_0,$$

so the output mean has magnitude bounded below along a subsequence by a nonzero constant times $|\mu|^k$. Jensen's inequality gives

$$|\mathbb{E}y_k|^2 \leq \mathbb{E}|y_k|^2.$$

Thus $\mathbb{E}|y_k|^2$ cannot decay asymptotically faster than $|\mu|^{2k}$. This rules out any uniform contraction factor below $|\mu|^2$. $\square$

Lemma 2.2 (A necessary determinant condition for stability). *If a real square matrix M satisfies $\rho(M) < 1$, then*

$$\det(I - M) > 0.$$

Proof. The eigenvalues of a real matrix are real or occur in conjugate pairs. If $|\lambda_i| < 1$, then each real factor $1 - \lambda_i$ is positive and each conjugate pair contributes $|1 - \lambda_i|^2 > 0$. Multiplying the factors gives $\det(I - M) > 0$. $\square$

The proof strategy is always the same. First impose mean-square stability on the high-curvature quadratic $\lambda = L$. By Lemma 2.2, this forces an algebraic restriction on the stepsize and momentum parameters. Then move to the low-curvature quadratic $\lambda = m$. Under the high-curvature restriction, the low-curvature mean transition has an eigenvalue too close to one; Lemma 2.1 turns that mean eigenvalue into a mean-square lower bound.

3 Nesterov's accelerated method

The NAM recursion is

$$x_k = y_k + \beta(y_k - y_{k-1}), \quad y_{k+1} = x_k - \eta g_k. \quad (3)$$

On f_λ , put $\alpha = \eta\lambda$. Since

$$x_k = (1 + \beta)y_k - \beta y_{k-1},$$

we have

$$y_{k+1} = (1 - \alpha\zeta_k)((1 + \beta)y_k - \beta y_{k-1}). \quad (4)$$

Let

$$q(\alpha) = \mathbb{E}(1 - \alpha\zeta_k)^2 = 1 - 2\alpha + (1 + \delta)\alpha^2, \quad r(\alpha) = \mathbb{E}(1 - \alpha\zeta_k) = 1 - \alpha.$$

For

$$U_k = (\mathbb{E}y_k^2, \mathbb{E}[y_k y_{k-1}], \mathbb{E}y_{k-1}^2)^\top,$$

(4) gives $U_{k+1} = M_N(\alpha, \beta, \delta)U_k$, where

$$M_N(\alpha, \beta, \delta) = \begin{pmatrix} q(1 + \beta)^2 & -2q\beta(1 + \beta) & q\beta^2 \\ r(1 + \beta) & -r\beta & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (5)$$

The mean transition is

$$T_N(\alpha, \beta) = \begin{pmatrix} (1 - \alpha)(1 + \beta) & -(1 - \alpha)\beta \\ 1 & 0 \end{pmatrix}. \quad (6)$$

3.1 The high-curvature restriction

Set

$$a = \eta L.$$

On the high-curvature quadratic $\lambda = L$, a direct expansion of (5) gives

$$\det(I - M_N(a, \beta, \delta)) = -aB_N(a, \beta, \delta), \quad (7)$$

where

$$B_N(a, \beta, \delta) = (1 + \delta)\beta(2\beta + 1)a^2 + \{-4\beta^2 + \beta(\delta - 1) + \delta + 1\}a + 2\beta^2 - 2. \quad (8)$$

Equivalently, with

$$b = 1 - \beta \in (0, 1],$$

$$\begin{aligned} B_N(a, 1 - b, \delta) &= (1 + \delta)(1 - b)(3 - 2b)a^2 \\ &\quad + a\{-4b^2 - \delta b + 9b + 2\delta - 4\} + 2b^2 - 4b. \end{aligned} \quad (9)$$

Lemma 3.1 (NAM high mode, $\delta > 2$). *If $\delta > 2$ and the high-curvature NAM second-moment recursion is stable, then*

$$a < \frac{2(1 - \beta)}{\delta - 2}.$$

Proof. By Lemma 2.2 and (7), stability implies

$$B_N(a, \beta, \delta) < 0.$$

As a polynomial in a , B_N is convex and

$$B_N(0, \beta, \delta) = 2\beta^2 - 2 \leq 0,$$

with strict inequality unless $\beta = 1$, which is outside the parameter range. Put $b = 1 - \beta$ and

$$A = \frac{2b}{\delta - 2}.$$

Substituting $a = A$ into (9) gives

$$B_N(A, 1 - b, \delta) = \frac{2b^2}{(\delta - 2)^2} ((4\delta + 4)b^2 - (14\delta + 2)b + 13\delta - 8). \quad (10)$$

The bracketed quadratic is decreasing on $[0, 1]$, because its derivative is at most

$$8\delta + 8 - (14\delta + 2) = 6(1 - \delta) < 0,$$

and its value at $b = 1$ is $3(\delta - 2) > 0$. Hence (10) is positive. Since the convex polynomial $B_N(\cdot, 1 - b, \delta)$ starts negative at zero and is positive at A , its positive root is smaller than A . Therefore $B_N(a, \beta, \delta) < 0$ forces $a < A$. $\square$

Lemma 3.2 (NAM high mode, $\delta = 2$). *Let $\delta = 2$ and $b = 1 - \beta$. If the high-curvature NAM second-moment recursion is stable, then either $b > 1/2$, or*

$$a \leq 2\sqrt{b}.$$

Proof. At $\delta = 2$, (9) becomes

$$B_N(a, 1 - b, 2) = 3(1 - b)(3 - 2b)a^2 + b(7 - 4b)a + 2b^2 - 4b.$$

If $b \leq 1/2$, then the quadratic coefficient is at least 3, the linear coefficient is nonnegative, and stability implies $B_N(a, 1 - b, 2) < 0$. Hence

$$3a^2 - 4b < 0,$$

which gives the claimed bound. The remaining case is precisely $b > 1/2$. $\square$

3.2 The low-curvature lower bound

On the low-curvature quadratic $\lambda = m$, the NAM mean matrix uses

$$\alpha = \eta m = \frac{a}{\kappa}.$$

With $b = 1 - \beta$, the characteristic polynomial of (6) is

$$p_N(z) = z^2 - (1 - \alpha)(2 - b)z + (1 - \alpha)(1 - b). \quad (11)$$

Its discriminant is

$$\Delta_N = (1 - \alpha)\{b^2 - \alpha(2 - b)^2\}. \quad (12)$$

Lemma 3.3 (NAM low mode under a supercritical high-mode restriction). *Fix $A < \infty$. If*

$$0 < \alpha \leq \frac{Ab}{\kappa},$$

then there is a constant $C(A) < \infty$ such that

$$\rho(T_N(\alpha, 1-b))^2 \geq 1 - \frac{C(A)}{\kappa}$$

for all sufficiently large κ , uniformly over $b \in (0, 1]$.

Proof. If $\Delta_N < 0$, then (12) gives

$$b^2 < \alpha(2-b)^2 \leq 4\alpha \leq \frac{4Ab}{\kappa}.$$

Thus $b \leq 4A/\kappa$. The two roots are complex conjugates and have squared modulus

$$(1-\alpha)(1-b) \geq 1-b-\alpha \geq 1 - \frac{4A+4A^2}{\kappa},$$

where the last step uses $\alpha \leq Ab/\kappa$ and $\kappa \geq 1$.

If $\Delta_N \geq 0$, let λ_+ be the larger real root. The quadratic formula gives

$$1 - \lambda_+ = \frac{b + \alpha(2-b) - \sqrt{(1-\alpha)\{b^2 - \alpha(2-b)^2\}}}{2}.$$

Using $\sqrt{(1-u)(1-v)} \geq 1-u-v$ for $u, v \in [0, 1]$, applied with $u = \alpha$ and $v = \alpha(2-b)^2/b^2$, gives

$$\begin{aligned} 1 - \lambda_+ &\leq \frac{\alpha(2-b) + b\alpha + b\alpha(2-b)^2/b^2}{2} \\ &\leq \frac{3\alpha}{b} \leq \frac{3A}{\kappa}. \end{aligned}$$

Therefore $\lambda_+^2 \geq 1 - 6A/\kappa$, after increasing the constant if necessary. The two cases prove the lemma. $\square$

Lemma 3.4 (NAM low mode at the critical value). *At $\delta = 2$, if the high-curvature NAM second-moment recursion is stable, then*

$$\rho(T_N(a/\kappa, \beta))^2 \geq 1 - C\kappa^{-2/3}$$

for a universal constant $C < \infty$ and all sufficiently large κ .

Proof. Let $b = 1 - \beta$. If $b > 1/2$, then the condition $B_N(a, 1-b, 2) < 0$ bounds a by an absolute constant: indeed the positive root of the displayed polynomial in Lemma 3.2 is uniformly bounded for $b \in [1/2, 1]$. Hence $\alpha = a/\kappa = O(\kappa^{-1})$, and the larger low-curvature root of (11) is $1 - O(\kappa^{-1})$, which is stronger than the claimed bound.

Now suppose $b \leq 1/2$. By Lemma 3.2, $a \leq 2\sqrt{b}$, so

$$\alpha = \frac{a}{\kappa} \leq \frac{2\sqrt{b}}{\kappa}.$$

If the low-curvature roots are complex, then (12) gives

$$b^2 < \alpha(2-b)^2 \leq 4\alpha \leq \frac{8\sqrt{b}}{\kappa},$$

so $b = O(\kappa^{-2/3})$. The squared modulus is then

$$(1-\alpha)(1-b) \geq 1 - O(\kappa^{-2/3}).$$

If the roots are real, let λ_+ be the larger root. The proof of Lemma 3.3 gives

$$1 - \lambda_+ \leq \frac{3\alpha}{b},$$

while the quadratic formula also gives the crude bound

$$1 - \lambda_+ \leq \frac{b}{2} + \alpha$$

by dropping the nonnegative square-root term. If $b \geq \kappa^{-2/3}$, then

$$\frac{3\alpha}{b} \leq \frac{6}{\kappa\sqrt{b}} \leq 6\kappa^{-2/3}.$$

If $b < \kappa^{-2/3}$, then

$$\frac{b}{2} + \alpha \leq O(\kappa^{-2/3}).$$

Thus $\lambda_+^2 \geq 1 - O(\kappa^{-2/3})$ in all cases. $\square$

Proposition 3.1 (NAM lower bound). *For every fixed $\delta > 2$, every NAM tuning is either mean-square unstable on the hard quadratic f_L , or on the hard quadratic f_m has output mean-square contraction factor at least*

$$1 - \frac{C_N(\delta)}{\kappa}.$$

At $\delta = 2$, the corresponding lower bound is

$$1 - C_N\kappa^{-2/3}.$$

Consequently NAM has no uniform accelerated mean-square bias guarantee of the form (2) for $\delta \geq 2$.

Proof. For $\delta > 2$, Lemma 3.1 gives

$$a \leq Ab, \quad A = \frac{2}{\delta - 2}.$$

On the low-curvature instance, $\alpha = a/\kappa \leq Ab/\kappa$. Therefore Lemma 3.3 gives

$$\rho(T_N(a/\kappa, \beta))^2 \geq 1 - \frac{C_N(\delta)}{\kappa}.$$

The eigenvector corresponding to the root near one has nonzero y -coordinate, so Lemma 2.1 transfers this to the output mean-square lower bound.

For $\delta = 2$, the same conclusion follows from Lemma 3.4. Since, for any fixed $c > 0$,

$$1 - C\kappa^{-1} > 1 - c\kappa^{-1/2} \quad \text{and} \quad 1 - C\kappa^{-2/3} > 1 - c\kappa^{-1/2}$$

for all sufficiently large κ , the lower bounds contradict any uniform accelerated contraction (2). $\square$

4 Robust momentum method

The RMM recursion is

$$x_k = y_k + \frac{\beta}{\eta} z_k, \quad z_{k+1} = \beta z_k - \frac{\eta}{L} g_k, \quad y_{k+1} = y_k + z_{k+1}. \quad (13)$$

Write $h = \eta$. On the high-curvature quadratic $\lambda = L$,

$$z_{k+1} = -h\zeta_k y_k + \beta(1 - \zeta_k)z_k, \quad y_{k+1} = (1 - h\zeta_k)y_k + \beta(1 - \zeta_k)z_k. \quad (14)$$

For

$$U_k = (\mathbb{E}y_k^2, \mathbb{E}[y_k z_k], \mathbb{E}z_k^2)^\top,$$

the second-moment transition matrix is

$$M_R(h, \beta, \delta) = \begin{pmatrix} 1 - 2h + (1 + \delta)h^2 & 2\beta h\delta & \beta^2\delta \\ -h + (1 + \delta)h^2 & 2\beta h\delta & \beta^2\delta \\ (1 + \delta)h^2 & 2\beta h\delta & \beta^2\delta \end{pmatrix}. \quad (15)$$

A direct determinant calculation gives

$$\det(I - M_R(h, \beta, \delta)) = -h\{2\beta^2\delta + h(1 + \delta + 2\beta\delta) - 2\}. \quad (16)$$

Thus high-curvature mean-square stability implies

$$2\beta^2\delta + h(1 + \delta + 2\beta\delta) < 2. \quad (17)$$

On the low-curvature quadratic $\lambda = m$, put $\theta = 1/\kappa$. The mean transition matrix is

$$T_R(\theta; h, \beta) = \begin{pmatrix} 1 - h\theta & \beta(1 - \theta) \\ -h\theta & \beta(1 - \theta) \end{pmatrix}, \quad (18)$$

with characteristic polynomial

$$p_R(z) = z^2 - \{1 + \beta - \theta(h + \beta)\}z + \beta(1 - \theta). \quad (19)$$

Lemma 4.1 (RMM low mode, parameters bounded away from $\beta = 1$). *Fix constants $b_0, H > 0$. If*

$$0 \leq \beta \leq 1 - b_0, \quad 0 < h \leq H,$$

then, for all sufficiently small θ ,

$$\rho(T_R(\theta; h, \beta))^2 \geq 1 - C\theta,$$

where $C < \infty$ depends only on b_0 and H .

Proof. Write $b = 1 - \beta$, so $b \geq b_0$. We have

$$p_R(1) = \theta h \geq 0.$$

For a constant $K > H/b_0$, substitution into (19) gives

$$p_R(1 - K\theta) = \theta\{K^2\theta + Kb\theta - Kb - Kh\theta - K\theta + h\}. \quad (20)$$

Choose, for instance, $K = 2H/b_0 + 2$. Then $h - Kb \leq H - Kb_0 < 0$. The remaining terms in braces in (20) are $O_{b_0, H}(\theta)$, so for all sufficiently small θ , uniformly over the allowed parameters,

$$p_R(1 - K\theta) < 0.$$

Since the polynomial is positive at $z = 1$ and negative at $z = 1 - K\theta$, it has a real root in $[1 - K\theta, 1]$. Hence $\rho(T_R)^2 \geq (1 - K\theta)^2 \geq 1 - C\theta$. $\square$

Lemma 4.2 (RMM low mode at the critical high-mode boundary). *If $0 \leq \beta < 1$ and $0 < h \leq 1 - \beta$, then*

$$\rho(T_R(\theta; h, \beta))^2 \geq 1 - C\theta$$

for a universal constant $C < \infty$ and all sufficiently small θ .

Proof. Let $b = 1 - \beta$. If $b < 4\theta$, then the product of the two roots of (19) is $\beta(1 - \theta) = (1 - b)(1 - \theta)$. Whether the roots are real or complex conjugates, the squared spectral radius is at least this product. Therefore

$$\rho(T_R)^2 \geq (1 - b)(1 - \theta) \geq 1 - 5\theta.$$

Assume now $b \geq 4\theta$. Since $h \leq b$, direct substitution yields

$$\begin{aligned} p_R(1 - 2\theta) &= \theta(h - 2b) + \theta^2\{4 - 2(h + \beta)\} \\ &\leq -b\theta + 4\theta^2 \leq 0, \end{aligned}$$

while $p_R(1) = \theta h \geq 0$. Hence (19) has a real root in $[1 - 2\theta, 1]$. Thus

$$\rho(T_R)^2 \geq (1 - 2\theta)^2 \geq 1 - 4\theta.$$

The two cases prove the claim. $\square$

Proposition 4.1 (RMM lower bound). *For every fixed $\delta \geq 1$, every RMM tuning is either mean-square unstable on the hard quadratic f_L , or on the hard quadratic f_m has output mean-square contraction factor at least*

$$1 - \frac{C_R(\delta)}{\kappa}.$$

Consequently RMM has no uniform accelerated mean-square bias guarantee of the form (2) for $\delta \geq 1$.

Proof. If $\delta > 1$, (17) implies

$$\beta^2 \delta < 1, \quad \text{hence} \quad \beta \leq \frac{1}{\sqrt{\delta}} < 1.$$

It also implies $h \leq 2/(1 + \delta)$. Thus Lemma 4.1 applies with

$$b_0 = 1 - \frac{1}{\sqrt{\delta}}, \quad H = \frac{2}{1 + \delta}.$$

If $\delta = 1$, (17) becomes

$$2\beta^2 + 2h(1 + \beta) < 2,$$

or

$$h < 1 - \beta.$$

Then Lemma 4.2 applies. In both cases, Lemma 2.1 transfers the low-curvature mean eigenvalue to the output mean-square lower bound. The eigenvector issuing from the root at one when $\theta = 0$ has nonzero y -coordinate, and the same remains true for the nearby root for small θ . The lower bound $1 - C_R(\delta)/\kappa$ is incompatible with (2) for large κ . $\square$

5 Constant-ratio iDAM+

This section treats the constant-ratio regime

$$\tau = \frac{a_k}{A_k} \in (0, 1), \quad \frac{A_{k-1}}{A_k} = 1 - \tau.$$

On the quadratic $f_\lambda(x) = \lambda x^2/2$, write

$$s = \frac{\lambda}{m} \in [1, \kappa], \quad c = \tau^2 s.$$

Let

$$u_{k-1} := \nabla \phi_{k-1}^*(z_k).$$

The iDAM+ identities give

$$\nabla \phi_k^*(z_k) = \frac{u_{k-1} + \tau y_k}{1 + \tau}, \quad x_k = \frac{y_k + \tau u_{k-1}}{1 + \tau}. \quad (21)$$

Moreover,

$$\nabla \phi_k^*(z_{k+1}) = \nabla \phi_k^*(z_k) - \frac{a_k}{A_k m} g_k = \nabla \phi_k^*(z_k) - \tau s \zeta_k x_k.$$

Combining this with (21) yields the two-dimensional state recursion

$$\begin{pmatrix} y_{k+1} \\ u_k \end{pmatrix} = \frac{1}{1 + \tau} \begin{pmatrix} 1 - c \zeta_k & \tau(1 - c \zeta_k) \\ \tau(1 - s \zeta_k) & 1 - c \zeta_k \end{pmatrix} \begin{pmatrix} y_k \\ u_{k-1} \end{pmatrix}. \quad (22)$$

5.1 The high-curvature restriction

For the high-curvature quadratic, $s = \kappa$ and $c = \tau^2 \kappa$. Let $M_I(c, \tau, \delta)$ be the second-moment transition matrix induced by (22). A direct expansion gives

$$\det(I - M_I(c, \tau, \delta)) = -\frac{c}{(1 + \tau)^2} B_I(c, \tau, \delta), \quad (23)$$

where

$$\begin{aligned} B_I(c, \tau, \delta) &= (1 + \delta)(\tau^2 - 4\tau + 3)c^2 \\ &\quad + \{2\delta\tau + 2\delta - 2\tau^2 + 10\tau - 4\}c - 8\tau. \end{aligned} \quad (24)$$

For $0 < \tau < 1$, the coefficient of c^2 is positive.

Lemma 5.1 (iDAM+ high mode, $\delta > 2$). *If $\delta > 2$ and the high-curvature iDAM+ second-moment recursion is stable, then*

$$c < \frac{4\tau}{\delta - 2}.$$

Proof. By Lemma 2.2 and (23), stability implies

$$B_I(c, \tau, \delta) < 0.$$

The polynomial B_I is convex in c and $B_I(0, \tau, \delta) = -8\tau < 0$. Put

$$C = \frac{4\tau}{\delta - 2}.$$

Substituting into (24) gives

$$B_I(C, \tau, \delta) = \frac{8\tau^2}{(\delta-2)^2} [\delta^2 + \delta(2\tau^2 - 9\tau + 9) + 2\tau^2 - 6\tau - 4]. \quad (25)$$

The bracket in (25) is positive for $\delta > 2$ and $0 < \tau < 1$. Indeed, at $\delta = 2$ it equals

$$6(\tau - 1)(\tau - 3) > 0,$$

and its derivative with respect to δ is

$$2\delta + 2\tau^2 - 9\tau + 9 \geq 6$$

on $\delta \geq 2$, $0 < \tau < 1$. Hence $B_I(C, \tau, \delta) > 0$. Convexity then shows that $B_I(c, \tau, \delta) < 0$ forces $c < C$. $\square$

Lemma 5.2 (iDAM+ high mode, $\delta = 2$). *At $\delta = 2$, high-curvature stability implies*

$$c \leq C\sqrt{\tau}$$

for a universal constant $C < \infty$, once κ is sufficiently large.

Proof. At $\delta = 2$, (24) becomes

$$B_I(c, \tau, 2) = 3(\tau^2 - 4\tau + 3)c^2 + (14\tau - 2\tau^2)c - 8\tau.$$

If $\tau \leq 1/2$, then the quadratic coefficient is at least 3 and the linear coefficient is nonnegative. Stability implies $B_I(c, \tau, 2) < 0$, so

$$3c^2 - 8\tau < 0,$$

which gives $c \leq \sqrt{8\tau/3}$.

If $\tau > 1/2$, then $c = \tau^2\kappa \rightarrow \infty$ as $\kappa \rightarrow \infty$, while both the quadratic and linear coefficients in the displayed formula are positive. Hence $B_I(c, \tau, 2) > 0$ for all sufficiently large κ , contradicting stability. Thus every stable high-curvature tuning has $\tau \leq 1/2$ for large κ , and the first case applies. $\square$

5.2 The low-curvature lower bound

For the low-curvature quadratic, $s = 1$ and $c = \tau^2$. Taking expectations in (22) gives the deterministic mean matrix

$$T_I(\tau) = \frac{1}{1+\tau} \begin{pmatrix} 1-\tau^2 & \tau(1-\tau^2) \\ 0 & 1-\tau^2 \end{pmatrix}. \quad (26)$$

Both eigenvalues are exactly $1 - \tau$, and the corresponding visible eigenvector has nonzero y -coordinate.

Proposition 5.1 (Constant-ratio iDAM+ lower bound). *For every fixed $\delta > 2$, every constant-ratio iDAM+ tuning is either mean-square unstable on the hard quadratic f_L , or on the hard quadratic f_m has output mean-square contraction factor at least*

$$1 - \frac{C_I(\delta)}{\kappa}.$$

At $\delta = 2$, the corresponding lower bound is

$$1 - C_I\kappa^{-2/3}.$$

Consequently constant-ratio iDAM+ has no uniform accelerated mean-square bias guarantee of the form (2) for $\delta \geq 2$.

Proof. If $\delta > 2$, then [Lemma 5.1](#) gives

$$\tau^2 \kappa = c \leq \frac{4\tau}{\delta - 2}, \quad \text{hence} \quad \tau \leq \frac{4}{(\delta - 2)\kappa}.$$

On the low-curvature instance, [Lemma 2.1](#) and (26) give a mean-square lower bound of

$$(1 - \tau)^2 \geq 1 - \frac{8}{(\delta - 2)\kappa}.$$

At $\delta = 2$, [Lemma 5.2](#) gives

$$\tau^2 \kappa = c \leq C\sqrt{\tau},$$

so $\tau \leq C'\kappa^{-2/3}$. Therefore the low-curvature contraction factor is at least

$$(1 - \tau)^2 \geq 1 - C''\kappa^{-2/3}.$$

As in the NAM proof, these lower bounds are incompatible with $1 - c/\sqrt{\kappa}$ for any fixed $c > 0$ and sufficiently large κ . $\square$

6 Main theorem and remaining gaps

Combining [Propositions 3.1](#), [4.1](#) and [5.1](#) gives the promised partial lower bound.

Theorem 6.1 (Partial lower bounds for constant-parameter acceleration). *Under the growth condition (1) with $\sigma^2 = 0$, the one-dimensional multiplicative-noise oracle of [Section 2](#) gives the following impossibility results.*

- (a) *NAM with constant parameters $\eta > 0$, $0 \leq \beta < 1$, cannot guarantee a uniform accelerated mean-square bias contraction $1 - \Theta(\kappa^{-1/2})$ for $\delta \geq 2$.*
- (b) *RMM with constant parameters $\eta > 0$, $0 \leq \beta < 1$, cannot guarantee such a contraction for $\delta \geq 1$.*
- (c) *Constant-ratio iDAM+ with $0 < \tau < 1$ cannot guarantee such a contraction for $\delta \geq 2$.*

More quantitatively, the obstruction is $1 - O(\kappa^{-1})$ in the supercritical cases $\delta > 2$ for NAM/iDAM+ and $\delta \geq 1$ for RMM, and $1 - O(\kappa^{-2/3})$ at the critical value $\delta = 2$ for NAM and iDAM+.

Remark 6.1 (Why this is not a sharp-threshold theorem). The theorem proves only one side of a sharp-threshold statement. It does not show that NAM or iDAM+ accelerate for all $\delta < 2$, nor that RMM accelerates for all $\delta < 1$. The known sufficient analyses in [\[1\]](#) stop at $\delta < 1$ for NAM/iDAM+ and $\delta < 1/4$ for RMM. Therefore the intervals

$$1 \leq \delta < 2 \quad \text{for NAM and iDAM+}, \quad \frac{1}{4} \leq \delta < 1 \quad \text{for RMM}$$

remain unresolved by this argument.

Remark 6.2 (Why the construction is nevertheless relevant). The oracle is admissible under exactly the growth condition used in the problem: it is unbiased and has conditional variance $\delta \|\nabla f(x)\|^2$, with $\sigma^2 = 0$. The hard functions are also as simple as possible: two one-dimensional quadratics, one with curvature L and one with curvature m . The lower bound is therefore not an artifact of nonsmoothness, high dimension, biased noise, or additive variance.

References

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